The Standard Model from 600-Cell Closure Geometry:

Spectral Bridge, Structural Constants, and the φ-Permeability

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Abstract

The VFD closure programme has developed three pillars: spectral-geometric particle masses on the 600-cell (Papers I–V), structural correspondences to the Standard Model (Papers VI–XI), and a route from dissipative closure dynamics to Schrödinger-type evolution (Papers XII–XXI). This paper bridges them through computation and exhibits structural relations between 600-cell eigenvalue data and known physical constants.

An explicit H_4-invariant closure functional on the 600-cell is constructed. The complete spectral decomposition reveals that Paper V's mass eigenvalues $\{9, 12, 14, 15\}$ are exactly the nontrivial integer Laplacian eigenvalues — a number-theoretic selection principle. The $E_8 \to H_4$ Coxeter projection (exponents $1, 7$) maps all 240 E_8 roots onto 120 600-cell vertices (2 per vertex), establishing the double-cover structure.

From this structure, the paper identifies structural correspondences, including: $\alpha^{-1} = 137 + \pi/87$ at 0.81 ppm, where 137 is a graph invariant (87 triply overdetermined, 50 the eigenvalue trace) and $\pi/87$ is the Coxeter projection phase excess divided by the double cover; $\sin^2\theta_W(\text{GUT}) = 9/(9{+}15) = 3/8$, identified as the NN/NNN angular ratio $36°/60° = 3/5$; a three-level $\mathbb{Z}_3$ splitting suggestive of generation structure; and exact integer/irrational sector decoupling (94 physical modes, proved by representation theory). The golden ratio $\varphi = 2\cos 36°$ enters through the 600-cell distance structure: the permeability gap $d_2^2 - d_1^2 = 1/\varphi$ determines the tunneling scale from which all φ-power mass formulas arise.

Thirteen scripts verify every computational claim. The paper does not claim to derive the Standard Model from first principles; it presents evidence that a single closure functional on the 600-cell can coherently relate mass eigenvalues, gauge structure, fundamental constants, and generation structure, with gravity remaining as a separate programme.

1 Introduction

The VFD closure programme rests on three pillars:

1. **Spectral-geometric masses** (Papers I–V [Smart, 2026a,b]): thirteen non-reference particle masses at 0.014% average error from 600-cell Laplacian eigenvalues.

2. **Standard Model correspondence** (Papers VI–XI [Smart, 2026c,d]): gauge symmetry as projection invariance, mass as closure residual, confinement as connectivity constraint.

3. **Quantum dynamics** (Papers XII–XXI [Smart, 2026e,f,g,h,i]): Schrödinger evolution as the equilibrium tangent limit of nonlinear closure-paired dynamics, with Witten Hamiltonian H_{W}.

These pillars share the closure functional $\mathcal{F}$ but have not been formally connected. This paper constructs the bridges through explicit computation.

Scope and epistemic status

Results are classified at three levels throughout:

- **Computed/proved:** exact results from eigenvalue computation, representation theory, or algebraic identity. Verified by accompanying scripts.
- **Structural:** results that follow from the computed data under a specific physical interpretation (e.g. identifying eigenvalue ratios with coupling normalisations).
- **Open:** questions not resolved by this paper.

2 The Closure Functional

2.1 Explicit construction

Definition 1 (600-cell closure functional)**.** *Let $\{v_1, \ldots, v_{120}\} \subset \mathbb{R}^4$ be the 600-cell vertices on the unit 3-sphere. For smoothing parameter $\varepsilon > 0$:*

$$\mathcal{F}(x) = -\varepsilon^2 \log\left(\sum_{i=1}^{120} \exp\left(-\frac{|x - v_i|^2}{2\varepsilon^2}\right)\right) \tag{1}$$

Proposition 2 (Properties)**.** *$\mathcal{F}$ satisfies: (1) $\mathcal{F}(v_i)$ is constant on the vertex set and can be normalised to zero; for small ε, vertex values differ from the sharp-minimum limit only by exponentially small corrections; (2) $\mathcal{F}(x) > 0$ away from vertices; (3) H_4 invariance; (4) C^∞ smoothness; (5) uniform Hessian at each vertex (by vertex-transitivity); (6) barrier height between adjacent vertices controlled by ε.*

Proof sketch. (1): At $x = v_j$, the $i = j$ term dominates the sum, giving $\mathcal{F} \to 0$ as $\varepsilon \to 0$; for finite ε, corrections are exponentially small in $1/\varepsilon^2$. (2): Away from vertices, all $|x - v_i|^2 > 0$. (3): H_4 permutes the vertex set, leaving the sum invariant. (4): Log-sum-exp of C^∞ functions. (5): H_4 is transitive on vertices. (6): The barrier midpoint has $\mathcal{F} \approx d^2/(4\varepsilon^2)$ for $\varepsilon \ll d$. Verified computationally. □

2.2 Unit matching

The closure equation $i\partial_t\Psi = H_{\mathrm{W}}\Psi$ with $H_{\mathrm{W}} = -(\sigma^2/2)\Delta + V_{\mathrm{W}}$ matches the physical Schrödinger equation $i\partial_t\psi = -(\hbar/2m)\Delta\psi + (V/\hbar)\psi$ under:

$$\sigma^2 = \hbar/m, \qquad V_{\mathrm{W}} = V/\hbar \tag{2}$$

One physical energy scale fixes σ^2 completely.

3 The φ-Permeability

The golden ratio $\varphi = (1+\sqrt{5})/2 = 2\cos 36^\circ$ is the fundamental geometric constant of the 600-cell. It determines the vacuum field permeability — the tunneling scale between constraint-satisfying configurations.

3.1 Distance shells and angles

The 600-cell vertex separations on S^3 form eight distance shells. Every angular separation is a multiple of $\pi/5$ (the decagonal angle):

Table 1: Distance shells of the 600-cell. All angles are φ-determined.

Shell	Distance	Angle	Degree	cos(angle)
1	$1/\varphi$	$36^\circ = \pi/5$	12	$\varphi/2$
2	1	$60^\circ = \pi/3$	20	$1/2$
3	$\sqrt{3-\varphi}$	$72^\circ = 2\pi/5$	12	$1/(2\varphi)$
4	$\sqrt{2}$	$90^\circ = \pi/2$	30	0
5	φ	$108^\circ = 3\pi/5$	12	$-1/(2\varphi)$
6	$\sqrt{3}$	$120^\circ = 2\pi/3$	20	$-1/2$
7	$\sqrt{3+\varphi}$	$144^\circ = 4\pi/5$	12	$-\varphi/2$
8	2	$180^\circ = \pi$	1	-1

3.2 The fundamental permeability gap

Proposition 3 (Permeability gap)**.** *The squared-distance difference between the first two shells is:*

$$d_2^2 - d_1^2 = 1 - 1/\varphi^2 = 1/\varphi \tag{3}$$

Proof. $d_1^2 = 1/\varphi^2$ and $d_2^2 = 1$. Then $1-1/\varphi^2 = 1-(\varphi-1)/\varphi = 1/\varphi$, using $1/\varphi = \varphi-1$. □

This identity is the root of all φ-dependent physics: the WKB tunneling ratio between nearest and next-nearest neighbours is $t_2/t_1 = \exp(-C/\varphi\sigma^2)$, and mass ratios are exponentials of φ-weighted eigenvalue sums, producing the $\varphi^{(\mathrm{rational})}$ mass formulas of Papers I–V.

4 Spectral Decomposition

4.1 The tight-binding bridge

Proposition 4 (Tight-binding spectral bridge)**.** *In the tight-binding regime ($\sigma^2 \ll$ barrier height), the Witten Hamiltonian on the 600-cell landscape is:*

$$H_{\mathrm{W}}^{\mathrm{eff}} \approx E_0\,\mathbf{I} + t\,\mathbf{A}_{600} \tag{4}$$

where $\mathbf{A}_{600}$ is the adjacency matrix (uniform tunneling by vertex-transitivity).

4.2 Complete eigenvalue-representation map

The 120 vertices are the elements of the binary icosahedral group $2I$. The adjacency matrix, commuting with $2I$, is diagonal on irreducible representations with eigenvalue $\alpha_\rho = (\dim\rho)^{-1}\sum_{s\in S}\chi_\rho(s)$ where S is the 12-element nearest-neighbour set. Numerical diagonalisation confirms:

Table 2: Complete spectral decomposition (verified computationally).

$2I$ irrep	dim	mult	Adj. α	Exact	Lap. λ	Integer?
trivial	1	1	12	12	0	Yes
2	2	4	9.708	$3{+}3\sqrt{5}$	$9{-}3\sqrt{5}$	No
3	3	9	6.472	$2{+}2\sqrt{5}$	$10{-}2\sqrt{5}$	No
4	4	16	3	3	**9**	**Yes**
5	5	25	0	0	**12**	**Yes**
6	6	36	-2	-2	**14**	**Yes**
3′	3	9	-2.472	$2{-}2\sqrt{5}$	$10{+}2\sqrt{5}$	No
4′	4	16	-3	-3	**15**	**Yes**
2′	2	4	-3.708	$3{-}3\sqrt{5}$	$9{+}3\sqrt{5}$	No

Proposition 5 (Integer selection principle)**.** *Paper V's mass eigenvalues $\{9, 12, 14, 15\}$ are exactly the nontrivial integer Laplacian eigenvalues. The irrational eigenvalues (involving $\sqrt{5}$) are not used in the current particle assignments.*

4.3 Sector decoupling

Theorem 6 (Exact integer/irrational decoupling)**.** *Every distance-shell adjacency matrix $\mathbf{A}_d$ commutes with the $2I$ action and is therefore diagonal in the irrep basis. Since integer and irrational eigenvalue sectors occupy different irreps:*

$$P_{\mathrm{int}}\, f(\mathbf{A}_d)\, P_{\mathrm{irr}} = 0 \tag{5}$$

for every shell d and every function f. The 94-mode integer sector is exactly self-consistent.

This is an exact representation-theoretic result, not an approximation. Verified computationally for shells 1–3 and all powers of $\mathbf{A}$.

5 The E_8 Coxeter Projection

5.1 Projection construction

The Coxeter element of E_8 (order 30, verified) has eigenvalues $\exp(2\pi i m/30)$ at exponents $m \in \{1, 7, 11, 13, 17, 19, 23, 29\}$. The eigenvectors at exponents $(1, 7)$ define a 4-dimensional projection plane. This maps all 240 E_8 roots onto exactly 120 directions with 2 roots per direction, matching all 120 to 600-cell vertices (verified computationally).

5.2 The double cover

The ratio $240/120 = 2$ establishes the E_8 double cover of the 600-cell: two conjugate copies (Table 3).

Table 3: The conjugate pair within E_8.

Property	Copy A	Copy B
Spinor sector	$\lambda = 9$ (matter)	$\lambda = 15$ (antimatter)
NNN eigenvalue	-5	$+5$
Coxeter exponents	lower $\{1, 7, 11, 13\}$	upper $\{17, 19, 23, 29\}$
Galois character	$+\sqrt{5}$ terms	$-\sqrt{5}$ terms

5.3 Symmetry inheritance

Proposition 7. *If $\mathcal{F}$ is G-invariant, then H_{W} commutes with G and its eigenspaces decompose into G-representations.*

The H_4 symmetry of the 600-cell, connected to E_8 through the Coxeter projection and to $SU(5)$ through the GUT embedding chain, provides a candidate route for Standard Model gauge structure.

6 Structural Constants

6.1 The fine structure constant

Tree level: $\alpha_{\mathrm{tree}}^{-1} = 137$

Proposition 8. $(\lambda_1 - 1)(\lambda_2 - 1) - 1 + \sum_{i=1}^{4} \lambda_i = 87 + 50 = 137$ *for* $\{\lambda_i\} = \{9, 12, 14, 15\}$.

The number 87 is triply overdetermined:

1. $(9-1)(12-1) - 1 = 8 \times 11 - 1 = 87$ (mass eigenvalue algebra),
2. $3(14+15) = 3 \times 29 = 87$ (Paper V decomposition),
3. $\sum(\text{upper Coxeter exponents}) - 1 = (17+19+23+29) - 1 = 87$ (E_8 group theory).

Remark 9. *The identity $\sum(\text{upper exponents}) = 88 = (9-1)(12-1)$ connects the E_8 Coxeter structure directly to the mass eigenvalue algebra. The total exponent sum is $\sum m_i = 120$ (the vertex count).*

Structural phase correction: $\pi/87$

The Coxeter cycle accumulates total phase $2\pi \times 88$ in the upper (complementary) sector. Dividing by 2 for the double cover (240/120, computed) gives $\pi \times 88$ per 600-cell. Per degree of freedom ($87 = 88 - 1$):

$$\text{phase per d.o.f.} = \frac{\pi \times 88}{87} = \pi + \frac{\pi}{87} \tag{6}$$

The excess over the natural half-turn π is $\pi/87$. Every numerical factor has a structural origin: 88 from Coxeter exponents, 2 from the double cover, π from the phase geometry. However, the identification of this phase excess with the physical one-loop correction to α^{-1} remains a structural correspondence rather than a first-principles loop derivation.

RG consistency

The GUT coupling $\alpha_{\text{GUT}}^{-1} = 25 = \text{mult}(\lambda{=}12) = \dim(5)^2$ and $\sin^2\theta_W = 3/8$ give $\alpha_{\text{EM}}^{-1}(\text{GUT}) = 200/3$. One-loop SM running with 3 generations yields $\alpha^{-1} \approx 137$ at the Thomson limit, with $M_{\text{GUT}} \approx 5 \times 10^{14}$ GeV (within the standard range). The eigenvalue formula $87 + 50 + \pi/87 = 137.036$ (0.81 ppm) is the compact closed-form expression.

6.2 The Weinberg angle

Structural match: The eigenvalue ratio $\lambda_1/\lambda_4 = 9/15 = 3/5$ coincides with the $SU(5)$ hypercharge normalisation $\text{Tr}(Y^2)/\text{Tr}(T_3^2)$ (verified from the branching rules). Under $\sin^2\theta_W = (3/5)/(1 + 3/5) = 3/8$, this reproduces the GUT-scale Weinberg angle. It is also the angular ratio: NN angle / NNN angle $= 36°/60° = 3/5$. Whether this coincidence reflects a dynamical mechanism or is a structural correspondence of the eigenvalue algebra is not determined by the present analysis.

6.3 Coupling structure

Self-coupling ratios of the mass-carrying sectors: 9 : 0 : 4 : 9 (spinor : standard : highest : conjugate spinor). The spinor/conjugate-spinor equality (both = 9) reflects matter/antimatter symmetry. The cubic coupling $g_3 = 0$ by H_4 symmetry at vertices.

7 Generation Structure

The group $2I$ contains 20 elements of order 3, forming 10 conjugate $\mathbb{Z}_3$ subgroups. Under $\mathbb{Z}_3$, the spinor sector ($\lambda = 9$, multiplicity 16) decomposes as $8 \times \mathbf{1} + 4 \times \boldsymbol{\omega} + 4 \times \boldsymbol{\omega}^2$.

The 4 ω-states split under the closure Hamiltonian into **3 doubly-degenerate energy levels**: $E_1 \approx 6.24$, $E_2 \approx 2.25$, $E_3 \approx 0.46$ (computed). This three-level splitting structure is suggestive of a generation hierarchy, though the energy ratios ($E_1/E_2 \approx 2.8$, $E_2/E_3 \approx 4.9$) do not directly match the observed lepton mass ratios ($m_\tau/m_\mu \approx 17$, $m_\mu/m_e \approx 207$). The quantitative generation hierarchy requires the φ-tunneling structure of the mass formula (Section 8), not the raw $\mathbb{Z}_3$ energies.

A possible quantitative route to the generation mass hierarchy is through the φ-tunneling structure (Section 3) applied within the $\mathbb{Z}_3$ decomposition. Paper II's winding operator $f(w) = \varphi^5(w-1)^{1/\varphi}$ gives m_μ/m_e at 0.5% and m_τ/m_e at 3.7%.

8 The Mass Formula

The mass of a particle is determined by three φ-factors:

$$m = m_0 \times \varphi^{(\text{sector})} \times \varphi^{(\text{closure class})} \times \varphi^{(\text{generation})} \tag{7}$$

where:

- the *sector factor* comes from the integer eigenvalue ($\lambda \in \{9, 12, 14, 15\}$),
- the *closure-class factor* comes from the instanton action (with leading term $\Delta C = 15$, correction $50/81 = \sum \lambda_i / 3^4$),
- the *generation factor* comes from the $\mathbb{Z}_3$ energy splitting.

The exponent 1265/81 in $m_p/m_e = \varphi^{1265/81}$ decomposes as $15 + 50/81$, where 15 is the closure invariant ΔC and $81 = 3^4$ involves the spinor adjacency eigenvalue (3) raised to the spinor dimension (4). The instanton computation gives $S_{\text{NN}}/\sigma^2 = 0.501$, matching this structure.

The NN angular separation $36° = \pi/5$ is the fundamental decagonal angle of φ-geometry ($\varphi = 2\cos 36°$). The entire mass spectrum is built from the permeability scale set by φ through this angle.

9 UV Finiteness

The physical (integer) sector has 94 modes. The spectral zeta function $\zeta(2) = 1.74$ is finite. By Theorem 6, the irrational sector exactly decouples: the 94-mode truncation is an exact representation-theoretic result.

The target-space sector is therefore finite-dimensional, suggesting a built-in regularisation mechanism for the target-space degrees of freedom. This does not by itself constitute UV completeness in the field-theoretic sense: UV divergences in QFT arise from spatial momentum modes, which require separate treatment. If physical space emerges from $\mathcal{M}_{\text{cl}}$ (Papers IX–X), then $\mathcal{M}_{\text{cl}}$ is compact (S^3) with a discrete spatial spectrum, potentially giving full UV finiteness. Target-space mode finiteness is unconditional; spatial finiteness is conditional.

10 The Conjugate-Pair Interpretation

The E_8 root system contains two conjugate 600-cells (the double cover). Copy A carries matter ($\lambda = 9$, NNN $= -5$, lower Coxeter exponents). Copy B carries antimatter ($\lambda = 15$, NNN $= +5$, upper exponents). They are mirror partners under the Galois conjugation $\sqrt{5} \to -\sqrt{5}$.

One interpretation of the computed structure is that α measures the effective coupling between the conjugate sectors: $\alpha_{\text{tree}}^{-1} = 137$ is the algebraic distance in the eigenvalue structure; $\pi/87$ is the geometric phase from seeing one copy instead of the E_8 pair. The Weinberg angle ratio $9/(9+15) = 3/8$ would then be the balance point of this coupling.

This realises the crystallisation triplet of the flagship paper:

1. E_8 provides the **constraint space**,

2. the two conjugate 600-cells provide the **dual mechanisms**,

3. α provides the **selection coupling**.

The crystallisation operator $C[\Psi] = \operatorname{argmin} F[\Phi]$ admits a geometric realisation in terms of the Coxeter projection, with α as a candidate coupling constant.

Remark 10. *This interpretation is structural: it follows from the computed conjugate-pair structure and the role of α as an inter-sector coupling. It is not derived as a necessary consequence from the dynamics alone.*

11 Predictions and Falsifiability

The nonlinear closure residual $\delta U_{\mathrm{rel}} = \sigma^2(\Delta|\Psi|/|\Psi| - \Delta R_{\mathrm{st}}/R_{\mathrm{st}})$ from Papers XIX–XX is gauge-invariant, norm-preserving, and vanishes at equilibrium. It predicts nonlinear corrections to quantum mechanics far from equilibrium, with a structurally fixed coefficient. This is falsifiable: if experiment bounds the correction to zero, the framework reduces to standard QM.

The framework is also falsifiable through the mass programme (if the eigenvalue structure cannot reproduce further particles), the gauge projection (if the representation decomposition does not match SM quantum numbers), and the generation structure (if a mechanism excluding the 4th $\mathbb{Z}_3$ pair cannot be identified or a heavy 4th generation is excluded).

12 What Remains Open

1. **Gravity.** Papers IX–X identify $\mathcal{M}_{\mathrm{cl}}$ curvature as a gravitational analogue but do not derive the Einstein equations. This is a separate programme.

2. **Quantitative φ-mass derivation.** The mechanism ($d_2/d_1 = \varphi$, instanton structure) is identified; full barrier integrals for all particles are a defined further computation.

3. **Full $SU(5)$ branching.** The Coxeter projection and per-sector root content are computed; exact multiplicities after $SU(3)\times SU(2)\times U(1)$ branching require further refinement.

4. **Polytope uniqueness.** Whether the 600-cell is the unique polytope producing this structure is not proved.

13 Discussion

13.1 Computational reproducibility

Thirteen scripts in `scripts/` verify every claim: spectral verification, Coxeter projection, coupling structure, α derivation and one-loop structure, generation decomposition and selection, φ-mass connection, $SU(5)$ branching, deep structure, remaining gaps, medium gaps, mass and generations. All require only `numpy` and `scipy`.

13.2 What is computed and proved

- Integer selection principle (Proposition 5): Paper V eigenvalues = nontrivial integer Laplacian eigenvalues.
- Exact sector decoupling (Theorem 6): 94-mode truncation is representation-theoretically exact.
- $d_2/d_1 = \varphi$ exactly; permeability gap $d_2^2 - d_1^2 = 1/\varphi$ (Proposition 3).
- E_8 Coxeter projection: 120/120 vertex matching, 2 roots per vertex.
- Triple overdetermination of 87 (Proposition 8): three independent algebraic routes.
- $\mathbb{Z}_3$ energy splitting: 4 ω-states $\to$ 3 doubly-degenerate levels (computed).
- Coupling self-ratios $9 : 0 : 4 : 9$; $g_3 = 0$ by H_4 symmetry.

13.3 What is structurally identified (numerical identity computed, physical identification structural)

- $\alpha^{-1} = 137 + \pi/87$: the numerical decomposition is exact; the identification of $\pi/87$ as a physical one-loop correction is structural.
- $\sin^2\theta_W = 3/8$: the ratio $9/15 = 3/5$ is computed; its identification as the $SU(5)$ normalisation is a structural match.
- The generation mass hierarchy: 3 levels are computed; the quantitative mapping to $e/\mu/\tau$ masses requires the winding operator.

13.4 What is structural but not derived

- The identification of the 600-cell as the specific constraint geometry.
- The α formula as a compact closed-form expression numerically consistent with the RG running (one-loop gives ~ 137, the formula gives 137.036).
- The conjugate-pair interpretation of E_8 as crystallisation mechanism.
- The $\pi/87$ as Coxeter projection phase (computed, but the connection to the one-loop effective action is structural, not derived from a loop integral).

13.5 What is not claimed

- Derivation of the Standard Model from first principles.
- Full account of gravity.
- That the 600-cell is the unique geometry producing this structure.
- That the nonlinear closure residual has been experimentally tested.

14 Conclusion

This paper shows that a single closure functional $\mathcal{F}$ on the 600-cell constraint geometry can coherently relate: mass eigenvalues (through the graph Laplacian), gauge structure (through H_4/E_8 symmetry inheritance), the fine structure constant (137 from eigenvalue algebra, $\pi/87$ from the Coxeter double-cover phase), the Weinberg angle (from the $36°/60°$ angular ratio), generation structure (from $\mathbb{Z}_3$ energy splitting), and quantum dynamics (through the Witten Hamiltonian).

A central geometric mechanism in the framework is the φ-permeability of the 600-cell: the golden ratio $\varphi = 2\cos 36°$ determines the vertex geometry, and the permeability gap $d_2^2 - d_1^2 = 1/\varphi$ determines the tunneling scale from which all φ-power mass formulas, coupling ratios, and structural constants arise. The programme comes full circle: the crystallisation operator proposed in the flagship paper admits a geometric realisation in terms of the $E_8 \to$ 600-cell Coxeter projection, with α as a candidate coupling between the two conjugate copies.

Gravity remains as a separate programme. Thirteen scripts verify every computational claim independently.

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